LEVEL 16 · PART VIII

Advanced Kubernetes 16

Goal: Secure, extend and scale Kubernetes for multi-team production use.

RECOMMENDED TIME	TOPICS
4-6 weeks	10

PRACTICE / CAPSTONE

Harden a kind/EKS cluster: Pod Security Standards, default-deny NetworkPolicies, an admission policy, autoscaling, and a mesh with mTLS and canary routing.

Topics

#	Topic	What you will learn
1	RBAC in Depth **p.3139**	least-privilege roles, aggregated roles, audit, impersonation, common escalations
2	NetworkPolicies in Depth **p.3153**	default-deny patterns, egress control, Cilium/Calico, policy testing
3	Pod Security **p.3167**	Pod Security Standards/Admission, securityContext, seccomp, AppArmor
4	Operators **p.3179**	controller pattern, reconcile loops, real operators (Prometheus, Postgres), Operator SDK/kubebuilder
5	Custom Resource Definitions (CRDs) **p.3191**	defining CRDs, schemas, versions, writing a custom resource
6	Admission Controllers and Webhooks **p.3205**	mutating/validating, ValidatingAdmissionPolicy, Kyverno/OPA Gatekeeper
7	HPA, VPA and Cluster Autoscaler **p.3221**	pod vs node scaling, Karpenter, interactions and pitfalls
8	Stateful Workloads **p.3237**	running databases on K8s, operators, storage, backup, when not to
9	Service Mesh with Istio or Linkerd **p.3251**	architecture, sidecar injection, install and onboard an app

#	Topic	What you will learn
10	Mesh: mTLS, Routing, Retries, Timeouts, Circuit Breaking and Observability **p.3265**	VirtualService/DestinationRule or Linkerd equivalents, traffic shifting, golden metrics from the mesh
★	Level capstone project **p.3283**	Harden a kind/EKS cluster: Pod Security Standards, default-deny NetworkPolicies, an admission policy, autoscaling, and a mesh with mTLS and canary routing.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

